

Lunar Communication Scenario

1.0.1

Scenario definition

The Lunar Communication scenario has two service providers. The first provider manages 1 control centre, 1 relay control centre, 1 GS, 1 relay and 1 lunar asset. The second provider manages 1 rover control centre, 1 relay control centre, 2 GS, 1 relay and the lunar gateway able to communicate with all ground stations and relays as well as lunar assets. The other lunar relays provides local connectivity to further assets on the surface.

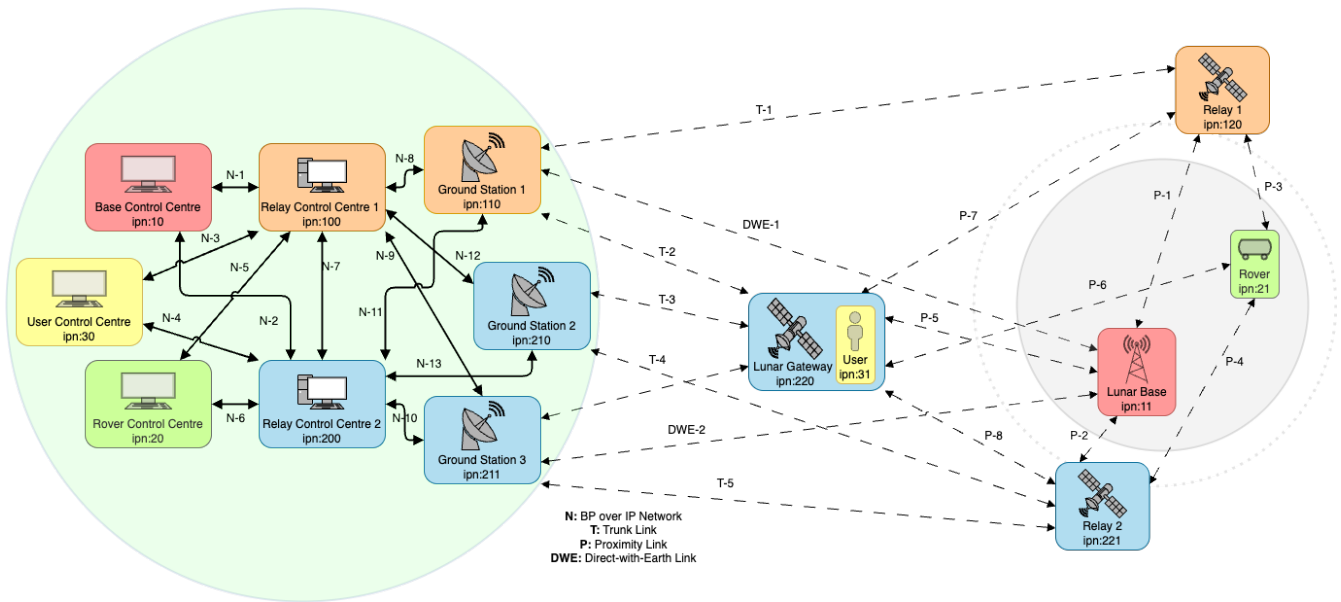
The orbit considered for the lunar gateway is realistic [1], while the two orbits of the Relays are approximation of a expected orbits for a future lunar constellation [2]. Also, the lunar assets position is according to expected future lunar exploration missions [3]. In Fig. 1 different assets of different organisational entities are represented with different colours and cross support is possible between the two entities managing the relay satellites and ground stations. The lunar base position is considering solar illumination and zones of interests for exploration [4]. The rover is placed on the far side of the moon so that the DTN capabilities can be properly tested in a sparsely connected environment. In this scenario we have a variety of link throughput and latencies. Terrestrial IP network in this scenario has a maximum throughput of 100 Mbps, the trunk links have a 100 Mbps downlink and 30 Mbps uplink. For lunar assets maximum throughputs are harder to define as different sources provide different capabilities, from 30 Mbps down to 2 Mbps for the rover and lunar base uplink and downlink.

Topology

The scenario has 13 DTN nodes separated in 5 different domains (colour coded the [image](#)):

- In red, lunar south pole base and a base control centre.
- In green, a rover on the lunar far side and a rover control centre.
- In Yellow, a user on board of the Lunar Gateway and a user control centre.
- In Orange, a service provider that controls a relay control centre, a ground station and a lunar relay satellite.
- In Blue, a service provider that control a relay control centre, two ground station, a lunar relay satellite and the lunar gateway.

It is important to note that the rover is on the far side of the moon and has no direct with Earth link connectivity. The lunar base position has been selected so that DWE connectivity is available, this is a desirable requirement for future exploration activity on the moon south pole.



Ground Assets

Ground Asset id	Ground Asset name	Cartesian X	Cartesian Y	Cartesian Z	Reference Center	Reference Axis	Notes
lgs1	Ground Station 1	-2414.0674 201 km	4907.86945 97 km	-3270.6048 732 km	Earth	ITRF	ESTRACK New Norcia
lgs2	Ground Station 2	4849.52524 63721 km	-360.60608 8513 km	4114.49510 06 km	Earth	ITRF	DSN Madrid
lgs3	Ground Station 3	-2351.1126 626803 km	-4655.5306 184282 km	3660.91268 75 km	Earth	ITRF	DSN Goldstone
lrov	Lunar Rover	-1737.4 km	0.0 km	0.0 km	Moon	MoonIAU2 009	Rover on the far side
lb	Lunar Base	-13.0 km	-12.47 km	-1737.4 km	Moon	MoonIAU2 009	Lunar Base on the South Pole

Orbital Parameters

Flying Asset id	Flying Asset name	Center of Orbit	Orbital Axes	Semi-major axes	Eccentricity	Inclination	Right Ascension	Argument of pericentre	True anomaly at reference epoch	Notes
Ire1	Lunar Relay 1	Moon	MoonIA U2009 frozen at J2000	9750.7 km	0.7	63.2 deg	0 deg	90 deg	0 deg	Moonlight Navigation Services, see reference [2]
Ire2	Lunar Relay 2	Moon	MoonIA U2009 frozen at J2000	9750.7 km	0.7	63.2 deg	120 deg	90 deg	164 deg	Moonlight Navigation Services, see reference [2]
Igw	Lunar Gateway	see reference [1]	see reference [1]	see reference [1]	see reference [1]	see reference [1]	see reference [1]	see reference [1]	see reference [1]	Lunar Gateway

Note: For the contact planning of the relays (except the Lunar Gateway) Keplerian dynamics has been assumed for the propagation of the orbits with the above parameters. Note: The reference epoch is 2026-01-01T00:00:00.00.

Data Rates

source\destination	Base Control Centre	User Control Centre	Rover Control Centre	Relay Control Centre 1	Relay Control Centre 2	Ground Station 1	Ground Station 2	Ground Station 3	Gateway	Relay 1	Relay 2	Rover	Lunar Base
Base Control Centre	-	0	0	100 Mbps	100 Mbps	0	0	0	0	0	0	0	0

source\destination	Base Control Centre	User Control Centre	Rover Control Centre	Relay Control Centre 1	Relay Control Centre 2	Ground Station 1	Ground Station 2	Ground Station 3	Gateway	Relay 1	Relay 2	Rover	Lunar Base
User Control Centre	0	-	0	100 Mbps	100 Mbps	0	0	0	0	0	0	0	0
Rover Control Centre	0	0	-	100 Mbps	100 Mbps	0	0	0	0	0	0	0	0
Relay Control Centre 1	100 Mbps	100 Mbps	100 Mbps	-	100 Mbps	100 Mbps	100 Mbps	100 Mbps	0	0	0	0	0
Relay Control Centre 2	100 Mbps	100 Mbps	100 Mbps	100 Mbps	-	100 Mbps	100 Mbps	100 Mbps	0	0	0	0	0
Ground Station 1	0	0	0	100 Mbps	100 Mbps	-	0	0	30 Mbps	30 Mbps	0	0	5 Mbps
Ground Station 2	0	0	0	100 Mbps	100 Mbps	0	-	0	30 Mbps	0	30 Mbps	0	5 Mbps
Ground Station 3	0	0	0	100 Mbps	100 Mbps	0	0	-	30 Mbps	0	30 Mbps	0	5 Mbps
Gateway	0	0	0	0	0	100 Mbps	100 Mbps	100 Mbps	-	15 Mbps	15 Mbps	100 Mbps	100 Mbps
Relay 1	0	0	0	0	0	100 Mbps	100 Mbps	100 Mbps	15 Mbps	-	15 Mbps	100 Mbps	100 Mbps

source\destination	Base Control Centre	User Control Centre	Rover Control Centre	Relay Control Centre 1	Relay Control Centre 2	Ground Station 1	Ground Station 2	Ground Station 3	Gateway	Relay 1	Relay 2	Rover	Lunar Base
Relay 2	0	0	0	0	0	100 Mbps	100 Mbps	100 Mbps	15 Mbps	15 Mbps	-	100 Mbps	100 Mbps
Rover	0	0	0	0	0	0	0	0	15 Mbps	15 Mbps	15 Mbps	-	0
Lunar Base	0	0	0	0	0	25 Mbps	25 Mbps	25 Mbps	15 Mbps	15 Mbps	15 Mbps	0	-

NOTE

The source of the data is on the left column and the destination on the upper row. The link are unidirectional and the table is not diagonally symmetric, thus reading the table from the top row may induce to a wrong value reading. The correct value should be read from the left column with the source as the starting reference.

- Every terrestrial link has a delay of 150ms.
- Every on board asset has a delay based on the OWLT computed at the distance of the assets at the beginning of the pass plus an additional 50ms processing delay.

Traffic Types and Volumes

Every flying and lunar asset produce TM&TC:

- Telecommand (TC) for commands to the spacecraft: 512 bytes of data every 5 minutes.
- Housekeeping Telemetry (TM) for mointoring of the spacecraft: 64KB of data every minute.

The lunar base, rover and user on the Lunar Gateway also produce:

- Scientific data generated by the PAYLOAD for downlink: 64KB of data every 1 seconds, for about 230 MB of data every hour.

Communication Planning

The communication planning fo the lunar communication scenario used to obtain the *actual contacts* is built around a hard requirements for the lunar base and gateway, that requires 24/7 coverage.

- The lunar base requires 24/7 coverage from relays.
- The lunar gateway requires 24/7 coverage from ground.

In order to provide 24/7 coverage for the lunar gateway, three ground stations are required, the

planning optimise the station coverage with no overlaps between ground stations. In this scenario, the ground stations are considered as ground station complex, that can cover multiple downlink/uplink at the same time.

Planned contacts heuristics:

- 24/7 GSs coverage for the LGW, with 3 ground stations sharing the coverage.
- The GSs passes are optimised based on duration and availability, if two GSs provide overlapping coverage, the longest pass is considered.
- 24/7 relay coverage for the lunar base, provided mostly by the LGW, with visibility gaps covered by other relays.
- Normal relays (not lgw) only communicate with ground stations of the same service provider.
- Normal relays can only communicate with one ground station at the time.
- Lunar rover coverage between the two normal relays is provided considering only the longest pass when coverage is overlapping.
- A proximity link with the lgw is available for 30 minutes after a lunar rover pass to the service provider not managing the lgw, so that the relay can downlink the data received from the rover.
- Proximity links between relays is not planned apart from the above case.
- LGW does not cover the rover, as the orbit provides minimal pass time.

With the provided topology and orbits, the heuristics provide a simplified but realistic communication planning between all the assets.

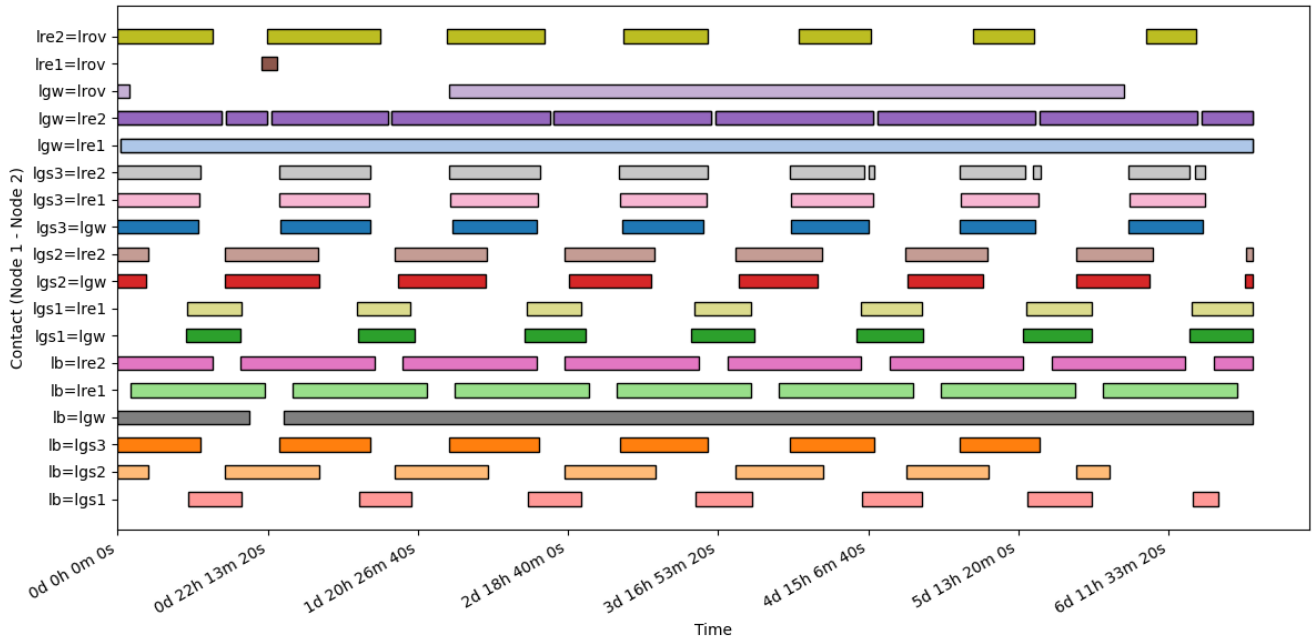


Figure 1. Asset visibility over 7 days

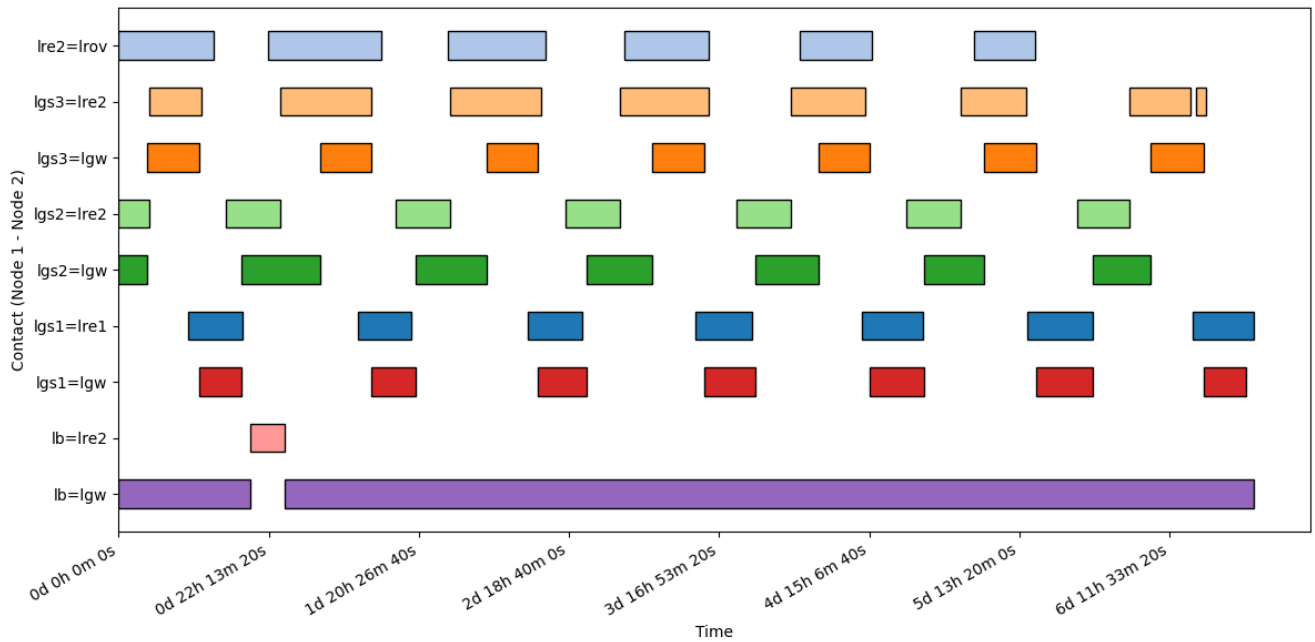


Figure 2. Planned Contacts links over 7 days

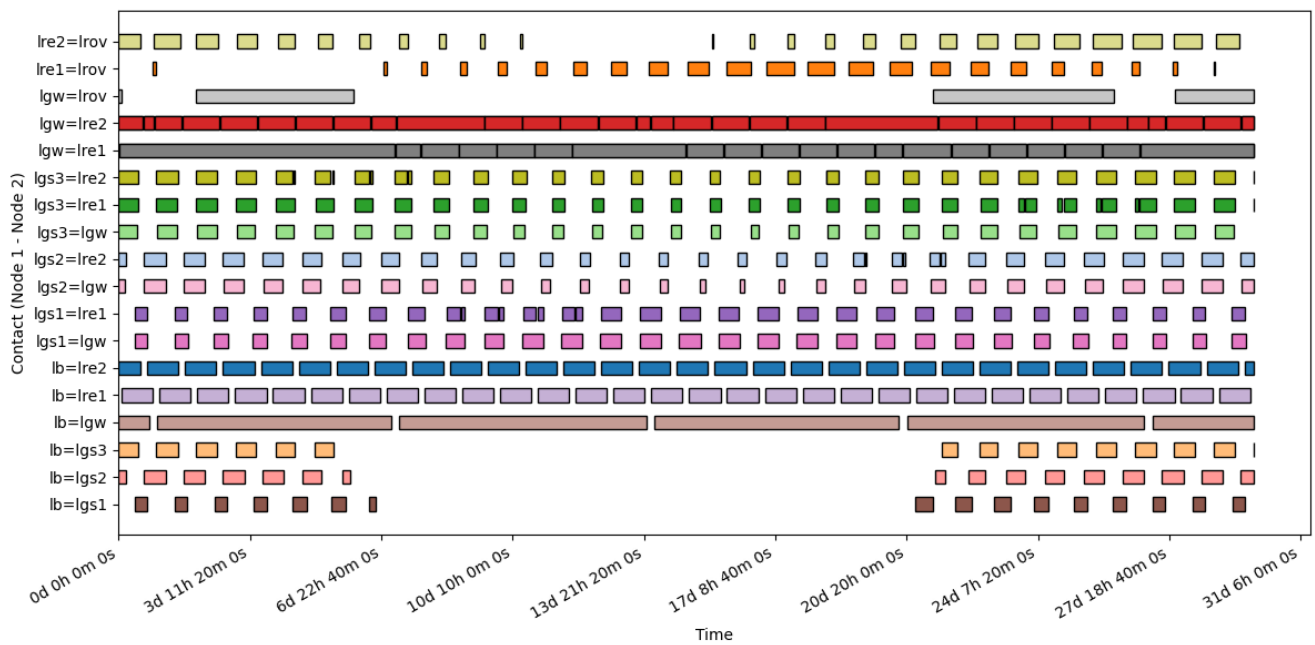


Figure 3. Asset visibility over 30 days

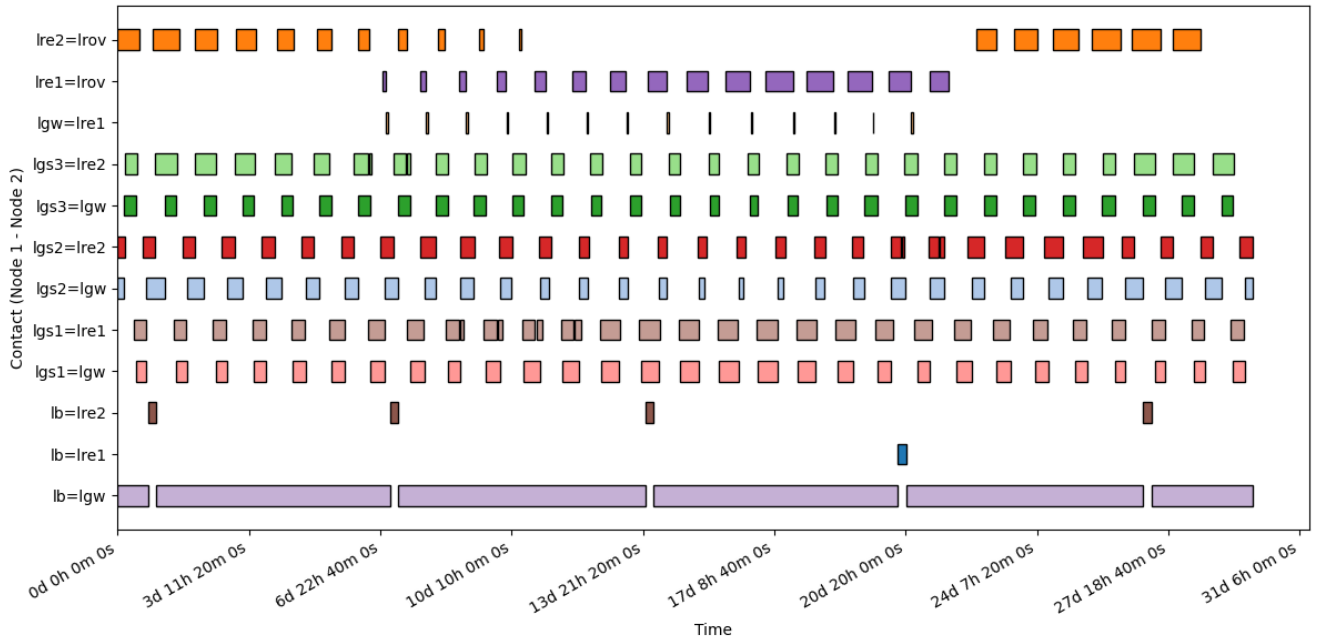


Figure 4. Planned Contacts links over 30 days

Error Model

In this scenario the error model is defined as:

- Probability of lost communication $P_l = 1\%$
- Probability of burst error during communication $P_b = 2\%$
 - A contact affected by burst error can have 1 to 10 errors of integer length between 1 and 5 seconds.
- Probability of delayed start or early end $P_d = 2\%$
 - The delay can last an integer number of minutes between 1 and 10, if the delay is longer than the length of the pass, the pass is dropped.
 - The early end can last an integer number of minutes between 1 and 5, if the early end is larger than the contact length, the pass is dropped.

The provided error model is worse than expected for routine lunar communication scenarios. It is provided to have a simple but interesting scenario. A user is expected to produce more realistic probabilities and produce an *actual contacts* file for more specific purposes or utilise a more sophisticated method to insert errors in the network.

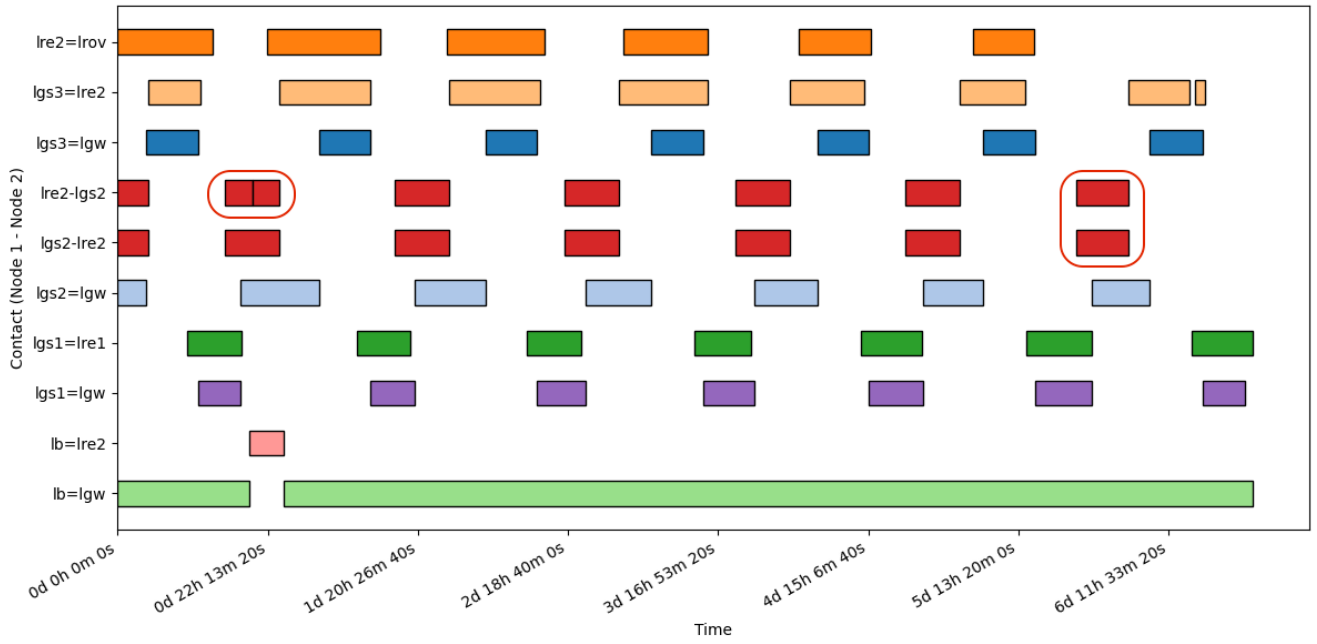


Figure 5. Actual Contacts links over 7 days

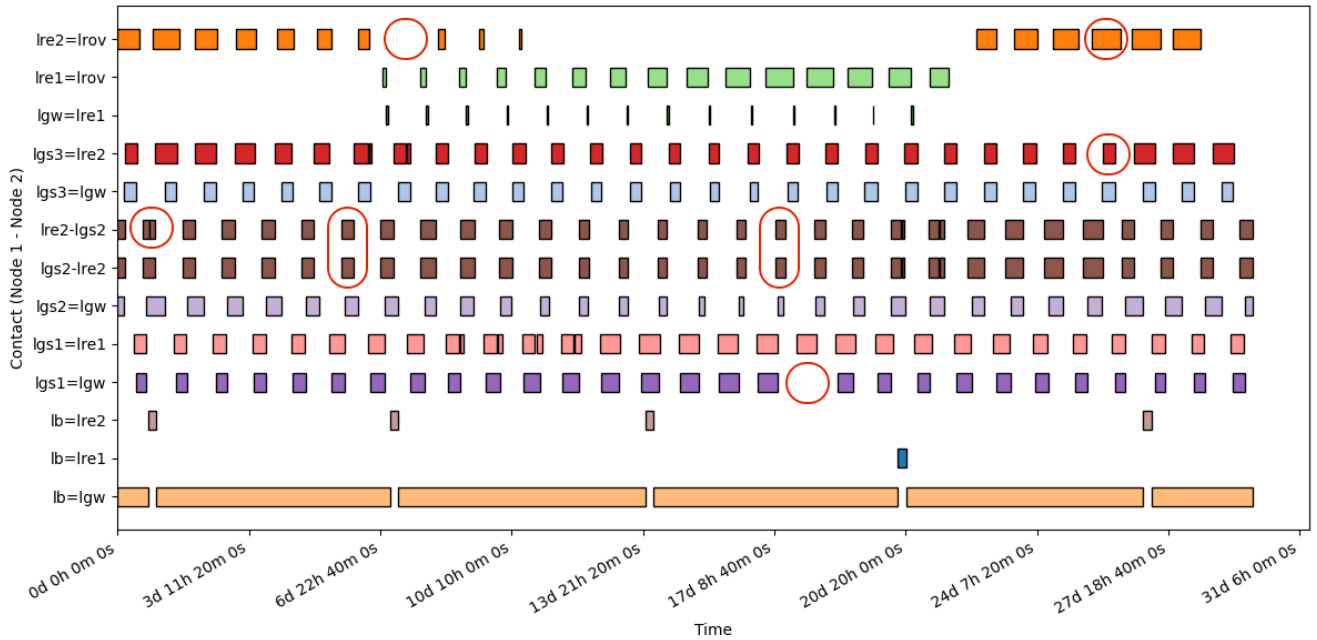


Figure 6. Actual Contacts links over 30 days

Note: In the *actual contacts* images, the link errors have been circled in red to highlight the differences with the *planned contacts*.

References

The following references have been used to produce the scenario.

- [1] Lunar Gateway Ephemeris, [Online]. Available: <https://naif.jpl.nasa.gov/naif/public.html>, https://naif.jpl.nasa.gov/pub/naif/misc/MORE_PROJECTS/DSG/
- [2] Giordano et al. (2021). Moonlight navigation service: How to land on peaks of eternal light, [Online]. Available:

- [3] Sathyan et al. (2024). Potential landing sites characterization on lunar south pole: De-Gerlache to Shackleton ridge region. *Icarus*, 412, [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0019103524000460>
- [4] Peña-Asensio et al. (2025). Evaluating potential landing sites for the Artemis III mission using a multi-criteria decision making approach. *Acta Astronautica*, 226, 469–478, [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0094576524006234> , https://www.researchgate.net/publication/358266257_Moonlight_navigation_service_-_how_to_land_on_peaks_of_eternal_light

Others:

- LunaNet Interoperability Specification Document, Version 5, [Online]. Available: <https://www.nasa.gov/wp-content/uploads/2025/02/lunanet-interoperability-specification-v5-baseline.pdf?emrc=606f95>
- Lunar Communication Architecture Working Group, IOAG, “The Future Lunar Communications Architecture”, V1.3, Jan. 31, 2022. [Online]. Available: <https://www.ioag.org/Public%20Documents/Lunar%20communications%20architecture%20study%20report%20FINAL%20v1.3.pdf>
- <https://logic.jhuapl.edu/> - Lunar Operating Guidelines for Infrastructure Consortium
- <https://www.ipnsig.org/> - Interplanetary Networking Special Interest Group (IPNSIG)